



Berkay Orhan

🌐 Nationality: Swedish
🏠 2002
📍 Norrköping, Sweden
🎓 Program: M.Sc Media Technology and Engineering
🗣️ Languages: *Swedish* Native, *Turkish* Native, *English* Advanced.

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PROJECTS

WikiLLM *LLM-Maintained Knowledge Bases* · Personal TypeScript CLI and agent plugin that turns raw project sources into cross-linked Obsidian wikis for Claude Code and Codex. Includes an npx vault wizard, generated AGENTS.md/CLAUDE.md schemas, ingest/query/lint skills, Marp export, and host-specific automation guidance.

FastTalk *Real-Time Voice Stack* · Personal Real-time voice conversation system with STT→LLM→TTS WebSocket streaming, Kokoro/Chatterbox TTS, PydanticAI tool calling, and AudioWorklet 48 kHz I/O. Built with Python, FastAPI, vLLM, faster-whisper, Docker, CUDA, Next.js, and React.

Nexus *Local Multi-Agent Harness* · Personal Local agent platform that routes one prompt through a classifier, orchestrator, and research/code/creative subagents in a sandboxed workspace. Includes MCP wrapper discovery, tiered provider routing, Claude OAuth/Codex CLI model paths, and a Next.js execution UI.

Oversee *Ops & AI Dashboard* · Personal Multi-tenant workshop platform with role-based access and an AI assistant for Excel parsing and report workflows. Includes human-in-the-loop approvals and Swedish-first analytics. Built with Next.js 16, Supabase, Vercel AI SDK, TanStack Table, and Playwright.

Stats for Spotify *Music Analytics Platform* · Personal Spotify analytics app for listening history, taste evolution, friend comparisons, in-browser playback, recap insights, and trend charts. Built with Next.js 16, Supabase, Spotify API, Recharts, and Row Level Security.

LiThePlan *University Course Planning* · Personal Course planning tool for Linköping University master's students with 339 courses, validated 90 hp study plans, drag-and-drop planning, conflict detection, and a Cmd+K palette. Built with Next.js 16, Supabase, Upstash Redis, and Playwright.

Kliv Idrottsförening *Sports Association Website* · Personal (Pro Bono) Website for a youth judo club with Google Calendar webhook sync, event admin, Nodemailer notifications, and an animated bento homepage. Built with Next.js 16, React, Google Calendar API, Redis, Clerk Auth, and Cloudinary.

Albyrådet *Non-Profit Web Application* · Personal (Pro Bono) Website for a youth organization in Alby with membership forms, Nodemailer contact flows, and a custom Framer Motion project carousel. Built with Next.js 15, Express.js, MongoDB, Mongoose, and shadcn/ui.

TECHNICAL SKILLS

Languages	TypeScript, Python, JavaScript, C++
Web/Fullstack	Next.js, React, Node.js, REST APIs, shadcn/ui, Tailwind CSS, Framer Motion, Radix UI
Backend/APIs	FastAPI, Express.js, WebSockets, Microservices
AI/ML/Agents	LangChain, LangGraph, Vercel AI SDK, Google Gemini, PyTorch, Whisper, CUDA, PydanticAI
Databases	Supabase, PostgreSQL, pgvector, Neo4j, MongoDB, Qdrant, Redis
Cloud/DevOps	Vercel, Google Cloud, Docker, Kubernetes
Specialized	Agentic RAG, GraphRAG, Multi-agent systems, MCP, Real-time STT/TTS/LLM streaming

EXPERIENCE

Jan 2026 – Jun 2026

Master's Thesis Student

Ericsson · Linköping

Designed and evaluated an agentic hybrid retrieval system for recommending relevant test cases in industrial CI/CD workflows. Combined dense semantic retrieval, sparse lexical matching, and knowledge-graph navigation within a ReAct-style agent workflow using LangChain, LangGraph, Neo4j, and pgvector.

Summer 2023 & 2024

Part-time Employee

LIDL · Stockholm

Performed various tasks including stocking shelves, assisting customers, operating the cash register, and maintaining cleanliness in the store.

EDUCATION

2024–Present

Master of Science in Media Technology & Engineering

Linköping University

Specializing in Machine Learning, AI, and Web Development.

2021–2024

Bachelor of Science in Media Technology & Engineering

Linköping University

Focus on programming, mathematics, and signal processing.

2019–2021

Naturvetenskapsprogrammet

Huddinge Gymnasium

2018–2019

Naturvetenskapsprogrammet

Tumba Gymnasium